



The k -hop connected dominating set problem: hardness and polyhedra[★]

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Abstract

Let $G = (V, E)$ be a connected graph, and k a positive integer. A subset $D \subseteq V$ is a k -hop connected dominating set (k -CDS) if the subgraph of G induced by D is connected and, for every vertex v in G , there is a vertex u in D such that the distance between v and u is at most k . We study the problem of finding a minimum k -hop connected dominating set, denoted by the acronym MINK-CDS . Firstly, we prove that MINK-CDS is NP-hard on planar bipartite graphs of maximum degree 4 and on planar biconnected graphs of maximum degree 5. We present an inapproximability threshold for MINK-CDS on bipartite and on $(1, 2)$ -split graphs, and we also prove that MINK-CDS is APX-hard on bipartite graphs of maximum degree 4. These results are shown to hold for every positive integer k . For $k = 1$, the classical minimum connected dominating set problem, we present an integer linear programming formulation and show some classes of inequalities that define facets of the corresponding polytope. We also present an approximation algorithm for this case.

Keywords: k -hop connected dominating set, hardness, polytope, facets

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1 Introduction

All graphs considered here are simple and undirected. In this section, all concepts and results refer to a connected graph $G = (V, E)$ with n vertices and maximum degree Δ ; and k is a positive integer. For every $u, v \in V$, we denote by $\text{dist}(u, v)$ the *distance* (the number of edges in a shortest path) between u and v . If v is a vertex in V and S is a nonempty subset of V , then $\text{dist}(v, S)$ denotes the distance from v to S , that is, $\text{dist}(v, S) = \min_{s \in S} \text{dist}(v, s)$. A subset $D \subseteq V$ is a *k-hop dominating set* (k -DS) of G if $\text{dist}(v, D) \leq k$ for every $v \in V$. A subset of vertices of G is a *k-hop connected dominating set* (k -CDS) if it is a k -DS and induces a connected subgraph. The sets 1-DS and 1-CDS are known as *dominating set* (DS) and *connected dominating set* (CDS).

Let $w: V \rightarrow \mathbb{Q}_+$ be a function that assigns nonnegative rational weights to the vertices of G . The *minimum weight k-hop connected dominating set problem* (MINK-WCDS) seeks for a k -CDS of minimum weight in a given vertex-weighted graph G . When $w(v) = 1$ for all $v \in V$, the problem is also known as the *minimum k-hop connected dominating set problem* (MINK-CDS). When $k = 1$, we write MINWCDS and MINCDS instead of MINK-WCDS and MINK-CDS. The problem MINK-WCDS has many applications in the design of wireless ad hoc and sensor networks [14].

It is known that MINCDS is $\mathcal{NP}$ -hard on split, chordal bipartite, and planar bipartite graphs; and it is also known to be solvable in polynomial time on strongly chordal, series-parallel and distance-hereditary graphs [9]. Nguyen and Huynh [11] showed that MINK-WCDS is $\mathcal{NP}$ -hard, for every k , on planar unit disk graphs with maximum degree 4.

In 2008, Du, Graham, Pardalos, Wan, Wu and Zhao [4] proposed an approximation algorithm for MINCDS with ratio $(1 + \ln(\Delta - 1))$. For MINWCDS, Guha and Khuller designed an approximation algorithm with ratio $1.35 \ln n$ [8]. As far as we know, these are the algorithms with the best approximation ratios described in the literature. Chlebík and Chlebíková [3] showed that, for every $\varepsilon > 0$, there is no $(1 - \varepsilon) \ln n$ -approximation algorithm for MINCDS on bipartite graphs and split graphs with n vertices, unless $\mathcal{NP} \subseteq \text{DTIME}(n^{O(\log \log n)})$. Recently, Bonsma [2] proved that MINCDS is $\mathcal{APX}$ -hard on cubic graphs.

Simonetti, Cunha and Lucena [12] proposed an integer linear programming formulation for MINWCDS with decision variables on the vertices and edges of the input graph. Their formulation uses an exponential number of constraints. In 2012, more compact formulations were proposed by Fan and Watson [6].

In this paper, we study algorithmic and polyhedral aspects of MINK-WCDS.

In Section 2, we show that Mink-CDS is $\mathcal{NP}$ -hard on planar bipartite graphs of maximum degree 4 and on planar biconnected graphs of maximum degree 5. Furthermore, we present an inapproximability threshold for Mink-CDS on bipartite and on $(1, 2)$ -split graphs. We also show that Mink-CDS is $\mathcal{APX}$ -hard on bipartite graphs of maximum degree 4. All these results are shown to hold for every positive integer k . In Section 3, we present an integer linear programming formulation that models MINWCDS . Moreover, we show inequalities that define facets of the polytope associated with this formulation. Finally, we present an approximation algorithm for MINWCDS based on rounding optimal solutions of the linear relaxation of our formulation. In the last section, we present some concluding remarks and discuss directions for further research. Owing to space limitation, proofs are omitted.

2 Hardness results for Mink-CDS

This section is devoted to $\mathcal{NP}$ -hardness and inapproximability results for Mink-CDS . As mentioned in the introduction, Nguyen and Huynh [11] showed that Mink-WCDS is $\mathcal{NP}$ -hard, for every k , on planar unit disk graphs with maximum degree 4. Their proof, which is quite involved, clearly implies that Mink-WCDS , for every k , is $\mathcal{NP}$ -hard on planar graphs with maximum degree 4. The next result, with a simpler proof, also implies the same fact, and, furthermore, it strengthens and generalizes a theorem of White, Farber and Pulleyblank [13], who showed that MINCDS is $\mathcal{NP}$ -hard on planar bipartite graphs. To prove Theorem 2.1, we show reductions from restricted versions of the minimum connected vertex cover problem (MINCVC) [7,5].

Theorem 2.1 *For every positive integer k , Mink-CDS is $\mathcal{NP}$ -hard on planar bipartite graphs of maximum degree 4, and on planar biconnected graphs of maximum degree 5.*

We state now an inapproximability result that we prove by showing an $\mathcal{AP}$ -reduction from the minimum set cover problem [1]. The subsequent $\mathcal{APX}$ -hardness result we obtain by an $\mathcal{L}$ -reduction from MINCVC [5].

Theorem 2.2 *For every positive integer k , Mink-CDS cannot be approximated within $c \ln n$ for a constant $c > 0$ on graphs with n vertices, unless $\mathcal{P} = \mathcal{NP}$. This result also holds on bipartite and on $(1, 2)$ -split graphs; and when $k = 1$, this holds for split graphs.*

Theorem 2.3 *For every positive integer k , Mink-CDS is $\mathcal{APX}$ -hard on bipartite graphs of maximum degree 4.*

3 Polyhedral results for MinWCDS

In this section, we present an integer linear programming formulation, which we call $\mathcal{F}$, that models MINWCDS, and study the facial structure of the associated polytope. First, we define some concepts. For a connected graph $G = (V, E)$, we say that a subset $\Gamma \subseteq V$ is a *separator* of G if $G[V \setminus \Gamma]$ is not connected. Define $\mathcal{S}(G)$ as the set of all inclusion-wise minimal separators of G . Let $\bar{\mathcal{C}}(G)$ denote the set of vertices of G that are not cut vertices, and let c be the number of cut vertices of G , that is, $c = n - |\bar{\mathcal{C}}(G)|$.

Kanté, Limouzy, Mary and Nourine [10] proved that a subset D of vertices of a graph G is a CDS if and only if $D \cap \Gamma \neq \emptyset$ for every $\Gamma \in \mathcal{S}(G)$. Thus, a cut vertex belongs to every CDS. Based on this characterization, it suffices to associate a decision variable $x_v \in \{0, 1\}$ only to vertices $v \in \bar{\mathcal{C}}(G)$. With the interpretation that $x_v = 1$ if and only if v belongs to a CDS of G , we obtain the following formulation.

$$\begin{aligned}
 (\mathcal{F}) \quad & \min \sum_{v \in \bar{\mathcal{C}}(G)} w(v)x_v \\
 & \text{s.t.} \quad \sum_{v \in \Gamma} x_v \geq 1 \quad \text{for all } \Gamma \in \mathcal{S}(G), |\Gamma| \geq 2 \quad (1) \\
 & \quad \quad x_v \in \{0, 1\} \quad \text{for all } v \in \bar{\mathcal{C}}(G). \quad (2)
 \end{aligned}$$

For every $D \subseteq V$, let $\chi(D) \in \{0, 1\}^{n-c}$ be the vector indexed by the vertices in $\bar{\mathcal{C}}(G)$ such that $\chi(D)_v = 1$ if and only if $v \in D \cap \bar{\mathcal{C}}(G)$. Let $\mathcal{P}(G) \subseteq \mathbb{R}^{n-c}$ be the polytope defined as $\text{conv}(\{\chi(D) : D \text{ is a CDS of } G\})$.

Proposition 3.1 *The following results hold for the polytope $\mathcal{P}(G)$.*

- (i) $\mathcal{P}(G) = \text{conv}(\{x \in \mathbb{R}^{n-c} : x \text{ satisfies (1) and (2)}\})$;
- (ii) $\mathcal{P}(G)$ is full-dimensional, that is, $\dim(\mathcal{P}(G)) = n - c$;
- (iii) For every $v \in \bar{\mathcal{C}}(G)$, the inequality $x_v \geq 0$ defines a facet of $\mathcal{P}(G)$ if and only if there is no $\Gamma \in \mathcal{S}(G)$ such that $v \in \Gamma$ and $|\Gamma| = 2$;
- (iv) For every $v \in \bar{\mathcal{C}}(G)$, the inequality $x_v \leq 1$ defines a facet of $\mathcal{P}(G)$.

For every subset $S \subseteq V$, let $N(S) = \{u \in V \setminus S : \{u, v\} \in E, v \in S\}$ be the *open neighborhood* of S in G . We say that a subset $S \subseteq V$ is *safe* if, for each connected component $\tilde{K}$ of $G - S$ and each cut vertex w of $G[\tilde{K}]$, we have $\bigcap_{i=1}^q N(K_i) \cap \Gamma \neq \emptyset$, where $K_1, \dots, K_q$ are the connected components of $G[\tilde{K}] - w$.

Proposition 3.2 *For every $\Gamma \in \mathcal{S}(G)$ with $|\Gamma| \geq 2$, the inequality $\sum_{v \in \Gamma} x_v \geq 1$ defines a facet of $\mathcal{P}(G)$ if and only if Γ is safe.*

The separation problem associated to inequalities (1) can be solved in polynomial time. More precisely, for graphs on n vertices, we can reduce it to $\mathcal{O}(n^2)$ instances of the maximum flow problem. Consequently, we obtain the following result, with which we derive the subsequent one.

Theorem 3.3 *The linear relaxation of $\mathcal{F}$ can be solved in polynomial time.*

Theorem 3.4 *There is a γ -approximation algorithm for MINWCDS, where γ is the size of the largest minimal separator of the input graph.*

4 Concluding remarks and further research

We showed that MINK-CDS is $\mathcal{NP}$ -hard on planar bipartite graphs of maximum degree 4. We also presented an approximation threshold for MINK-CDS on bipartite and on $(1,2)$ -split graphs, and we showed that MINK-CDS is $\mathcal{APX}$ -hard on bipartite graphs of maximum degree 4. All these results are for every positive integer k . To the best of our knowledge, the inapproximability results shown here for MINK-CDS , when $k \geq 2$, are the first to appear in the literature.

We presented an integer linear programming formulation for MINWCDS and showed some classes of inequalities that define facets of the polytope associated with this formulation. We are implementing a branch-and-cut algorithm based on these results, but we are still in an initial stage. Finally, we proved that there exists a γ -approximation algorithm for MINWCDS, where γ is the size of the largest minimal separator of the input graph. For graphs with n vertices and small separators, this algorithm may produce better solutions than the ones obtained with the $1.35 \ln n$ -approximation given by Guha and Khuller [8].

As our future steps, we intend to design approximation algorithms and to investigate the parameterized complexity of MINK-CDS , for $k \geq 2$. Additionally, we plan to extend our polyhedral studies to address MINK-WCDS .

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